

Qatar-1b

R-2744

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_190918 · created 2026-09-02T01:49:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458744.7036 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.051 +/- 0.056
Transit depth (Rp/Rs)^2	0.26 +/- 0.57 [%]
Orbital Inclination (inc)	84.96 +/- 2.89
Airmass coefficient 1 (a1)	1.09 +/- 0.51
Airmass coefficient 2 (a2)	-0.11 +/- 1.24
Scatter in the residuals of the lightcurve fit is	45.83 %
Best Comparison Star	#3 - [179, 29]
Optimal Aperture	0.87
Optimal Annulus	5.75
Transit Duration (day)	0.06 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.051 ± 0.056 vs 0.1463 (deviation 1.62σ)
Duration fit vs published	0.06 d vs 0.0692 d (deviation 0.46σ)
Mid-time O–C	-3.9 min
Depth SNR	0.9
Residual scatter	45.83 %

Light curve

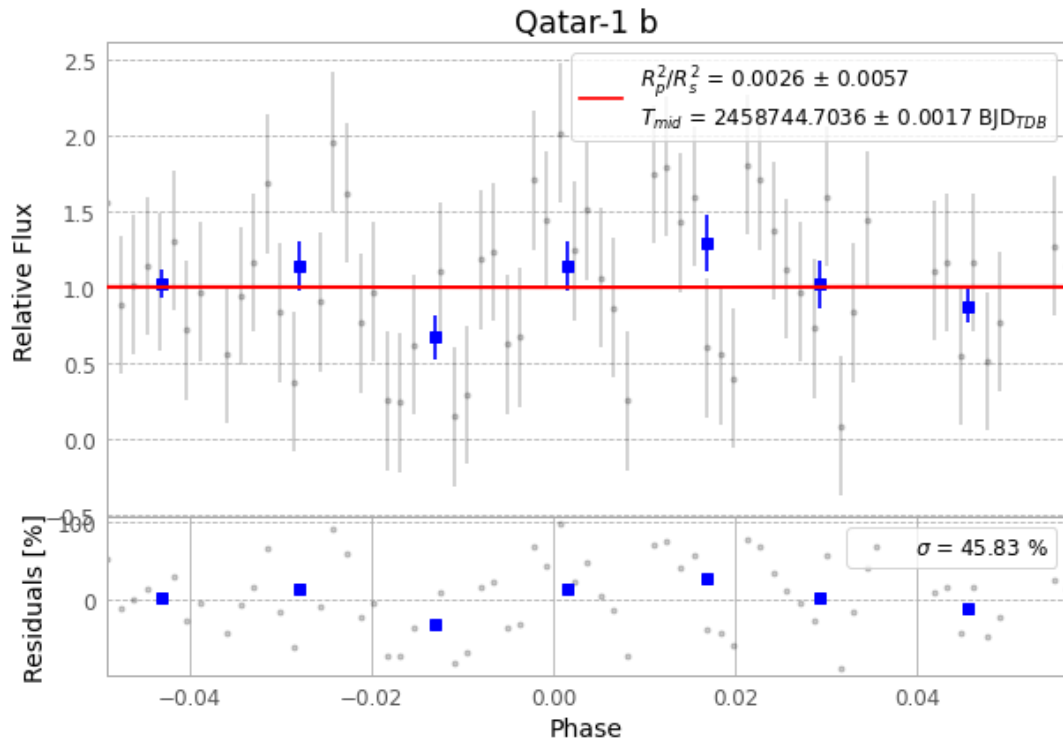


Figure 1 — FinalLightCurve_Qatar-1 b_2019-09-17.png (SHA-256 eda36b36f695c210...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [166, 309], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 06daafe78a1b3550...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1190918030912.FITS → Qatar-1190918064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190918.log (8cfd6bed857b...), FinalLightCurve_Qatar-1 b_2019-09-17.png (eda36b36f695...), FinalLightCurve_Qatar-1 b_2019-09-17.csv (fcbc2274268d...)

Compute resources

Wall time 196.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:49Z.